

FabricCo

Data Challenge 1 — Google Trends

Market Demand & Internal Performance Report

Team C · ISYS 814 · Dr. Shahrabi · April 2026

Executive Summary

Sophie needs to know whether swimsuits and scarves can sustain a dedicated manufacturing line year-round without switching. That question has two parts: what does external demand look like across the UK and NZ, and how does the existing internal portfolio perform? This report works through both.

The external picture is clear. Neither product has flat year-round demand, but they are seasonal in very different ways. Scarves spike hard and briefly in the winter months of each country, with volatility nearly three times that of swimsuits. Swimsuits build more gradually into summer and hold a broader plateau. Because the UK and NZ sit in opposite hemispheres, demand peaks arrive roughly six months apart. Whatever timing serves one market misses the other.

The internal picture is more nuanced. Five existing product lines are bunched within 8% of each other on unit volume, so volume cannot drive portfolio decisions. On a composite of six performance metrics, Bikinis leads, tied with Hoodies and Rain Coats at an average rank of 2.83. Beanies ranks last at 3.50, with the weakest gross margin (29.0%) and lowest customer satisfaction (8.41) in the portfolio. Three tested operational correlations sit at or near zero, meaning no single company-wide lever moves outcomes across all product lines.

Bottom line for Sophie and Andy: sequence a single scarf line and a single swimsuit line across UK and NZ seasons. The hemisphere offset makes year-round production viable without switching. Prioritize Bikinis for expansion. Review Beanies before scaling it further. Validate actual market volumes before committing manufacturing capacity.

Part 1: External Analysis — Google Trends

Google Trends outputs a relative search interest index from 0 to 100, normalized within each country and time window. Index levels cannot be compared across countries. Only shape and timing within a market are analytically valid. The combined dataset covers 268 monthly observations from January 2004 to the present: UK swimsuit, UK scarf, NZ swimsuit, and NZ scarf.

Step 2: UK — Swimsuits

UK swimsuit search interest is low and steady. The mean index across the full dataset is 10.5, the highest it has ever reached is 34, and the standard deviation is just 7.0. Interest picks up each spring, peaks in June and July, then fades through August. Before 2012, the series was nearly flat. Since then a slow upward drift is visible, but the overall level remains modest. For a manufacturer, this is a forgiving seasonal product: the summer build window is broad, roughly four months, and the timing does not need to be precise to the week.

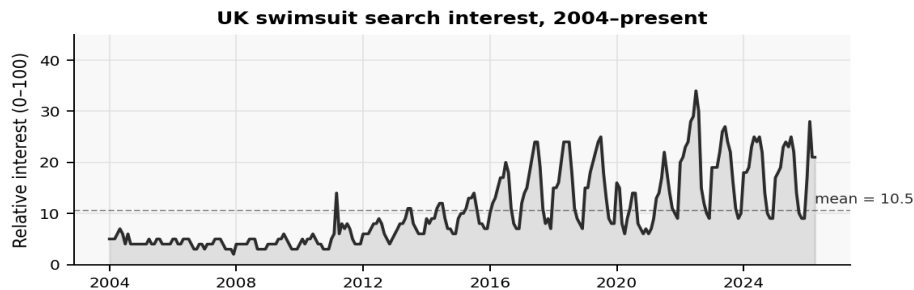


Figure 1 — UK swimsuit relative search interest, 2004–present. Mean = 10.5. Gradual summer peaks; low overall index.

Step 2: UK — Scarves

Scarves are a different problem. The mean index is 25.3, more than double swimsuits, and the series spikes to 100 in recent years. Every autumn, interest surges from a low base into a tall, narrow peak concentrated in October through December, then drops sharply into January. The spikes have intensified since around 2015. With a standard deviation of 20.4 versus swimsuit's 7.0, scarves are nearly three times more volatile. Miss the October build window and the season is gone.

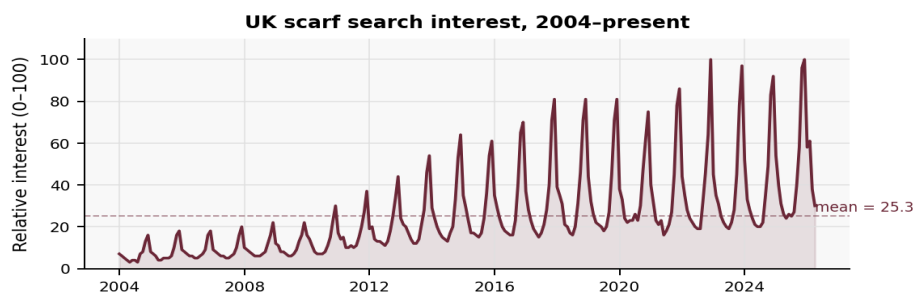


Figure 2 — UK scarf relative search interest, 2004–present. Mean = 25.3, std = 20.4. Sharp autumn–winter spikes, intensifying since 2015.

Step 3: New Zealand — Swimsuits

NZ swimsuit interest runs at a higher index than the UK within its own normalization (mean 24.6), and its seasonal shape is more sustained. Interest builds through November and holds through January before declining gradually. There is no sharp spike. The early years (2004–2006) show some noise, likely from low overall search volumes in the early Trends period; the pattern stabilizes from 2008 onward. Like the UK, NZ swimsuits offer a forgiving production window, roughly three months across the southern summer, without the time pressure that scarves impose.

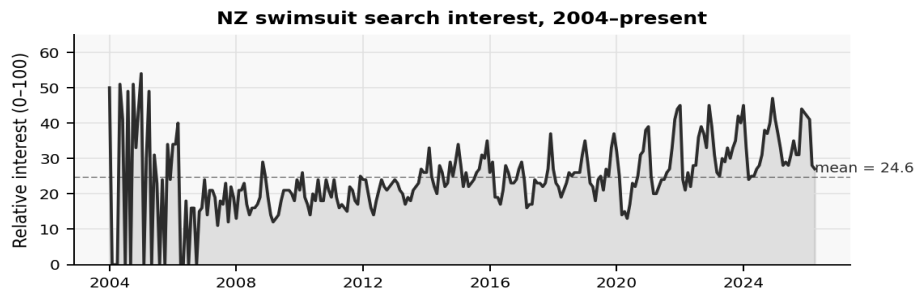


Figure 3 — NZ swimsuit relative search interest, 2004–present. Mean = 24.6. Broad southern-summer plateau (Nov–Jan).

Step 3: New Zealand — Scarves

NZ scarves mirror UK scarves almost exactly, rotated six months. Peaks arrive in May through July, have grown sharply in recent years, and reach 100 on the index. The standard deviation (20.8) is nearly identical to the UK's (20.4). Same spike height, same narrow window, different calendar. This near-perfect structural symmetry between the two markets' scarf patterns is the most important single finding in Part 1: the same product, the same risk profile, offset by exactly one hemisphere.



Figure 4 — NZ scarf relative search interest, 2004–present. Mean = 38.7, std = 20.8. Mirror image of the UK scarf pattern, six months later.

Step 3: Swimsuit–Scarf Relationship Within Each Country

Overlaying the two products on one axis makes the relationship visible. In both countries, they do not move together. When scarves are high, swimsuits tend to be low, and vice versa. The seasonal windows are not just different, they are largely opposite.

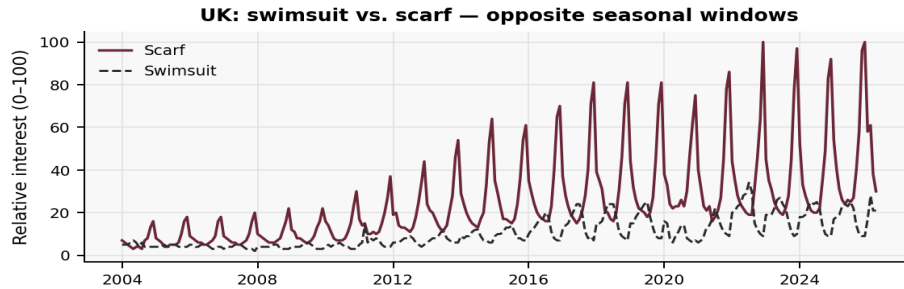


Figure 5 — UK: swimsuit vs. scarf overlaid. Opposite seasonal windows across the calendar year.

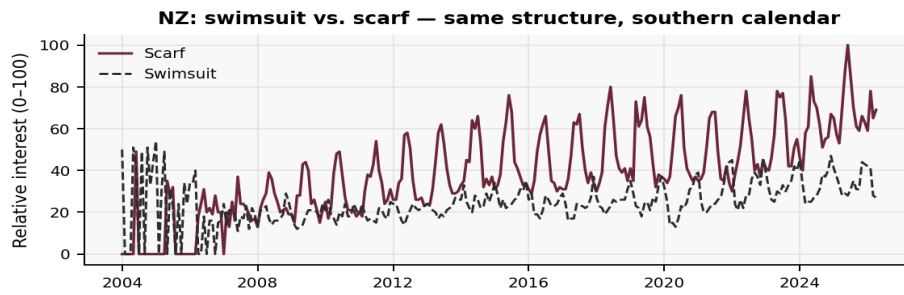


Figure 6 — NZ: swimsuit vs. scarf overlaid. Same inverse pattern on the southern calendar.

Pearson correlation puts a number on it: $r = 0.146$ in the UK and $r = 0.166$ in NZ. Both are weak positives, just above zero. The 0.02 difference between them is not meaningful.

Country	Pearson r	Swimsuit std	Scarf std	Relationship
UK	0.146	7.0	20.4	Weak — opposite seasonal windows
New Zealand	0.166	10.0	20.8	Marginally higher, still near-zero

Neither correlation is strong enough to use one product as a demand signal for the other. What the overlay charts do show, though, is something more useful: the two products are complementary within each market. One is rising while the other is flat. That complementarity becomes the strategic argument for carrying both.

Step 4: Cross-Country Comparison

Combining the UK and NZ datasets on a single time axis shows the hemisphere effect directly. The index levels cannot be compared across countries (each is normalized independently), but the timing is fully comparable and the offset is unambiguous.

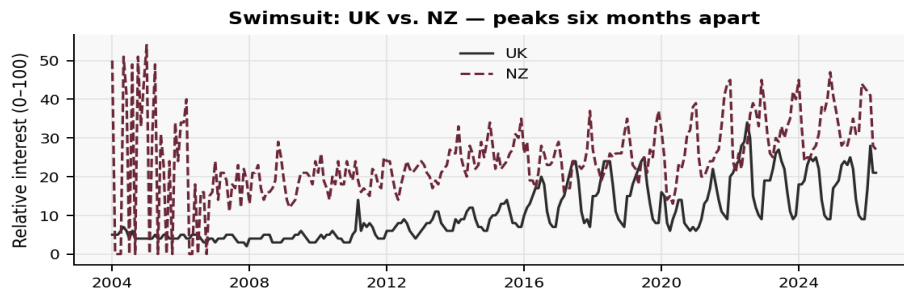


Figure 7 — Swimsuit: UK vs. NZ. UK peaks May–August; NZ peaks November–January. Six months apart, every year.

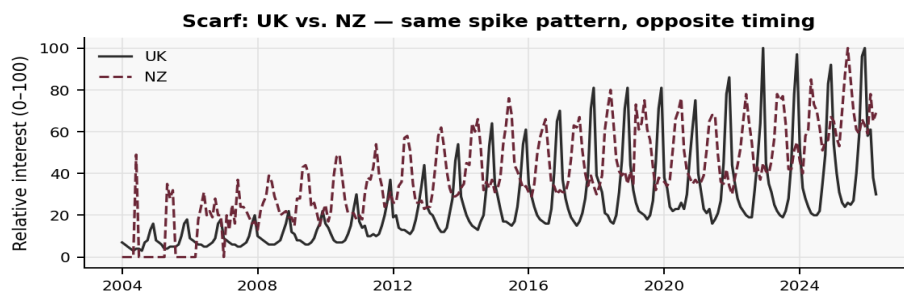


Figure 8 — Scarf: UK vs. NZ. UK peaks October–December; NZ peaks May–July. Same inversion.

The offset is consistent across both products and all 20 years of data. UK swimsuit peaks fall May–August without exception; NZ peaks fall November–January. UK scarf spikes hit October–December; NZ hits May–July. The two markets are almost exactly six months out of phase for both products.

The strategic implication follows directly: the UK and NZ markets are not two separate problems requiring two separate solutions. They are complementary halves of a continuous cycle. What is off-season in one market is in-season in the other, for both products, simultaneously.

Step 5: Implication for FabricCo — Manufacturing Lines

The answer to Sophie's question is yes. Year-round demand is achievable without switching manufacturing lines, provided FabricCo treats UK and NZ as a combined market rather than two independent ones.

A single scarf line can serve UK autumn demand (builds peaking around September) then shift to NZ autumn demand (builds peaking around March), with no calendar gap between them. A single swimsuit line runs UK summer builds March through June, then NZ summer builds September through December. Over a full year, both product lines have demand at virtually every point on the calendar.

Line	UK build window	NZ build window	Calendar gap
Scarves	Aug – Oct	Feb – Apr	None
Swimsuits	Mar – Jun	Sep – Dec	None

Two caveats. First, Google Trends measures search interest, not sales volume. The timing argument is solid; the absolute market size in each country still needs validation through sales data or retail research before capacity is committed. Second, the three-month build windows assume production lead times short enough to prepare for each market's season without holding large standing inventory. If lead times exceed the window, some forward buffering will be needed.

Prescription: sequence one scarf line and one swimsuit line across UK and NZ seasons using the hemisphere offset. Do not run parallel global lines. Validate demand volumes in each market before allocating manufacturing capacity.

Part 2: Internal Product Performance

The internal dataset covers Beanies, Bikinis, Hoodies, Summer Shirts, and Rain Coats across 252 months from January 2004 through December 2024, with ten KPIs per product per month.

Step 2: Exploratory Analysis

The first thing the data shows is how uniform the portfolio is. Every product sits between 342 and 356 mean monthly units, with standard deviations of 87 to 90. Margins cluster near 30%. Satisfaction runs from 8.42 to 8.60 on a 10-point scale. Inventory turnover falls between 5.4 and 5.6 turns per year. At this level of aggregation, the five products look nearly identical. That is not a trivial observation: it means any strategic differentiation has to come from the narrower gaps in margin, satisfaction, and efficiency rather than headline volume. Time-series charts showing trends for each of these four metrics across 2004–2024 are included in the Appendix (Figures A1–A4), generated from the full monthly dataset.

Product	Mean Units/Mo	Mean Margin	Mean Satisfaction	Mean Turnover
Beanies	355.5	29.8%	8.57	5.51
Bikinis	342.6	30.3%	8.54	5.51
Hoodies	350.6	29.6%	8.42	5.48
Summer Shirts	355.3	30.1%	8.42	5.43
Rain Coats	349.7	30.3%	8.60	5.56

Step 3: Comparative Performance (2020–2024)

Beanies leads on five-year unit volume at 21,994 units. Bikinis is last at 20,373. The spread between them is under 8%. Volume does not differentiate this portfolio and should not be the primary input to any resource allocation decision.

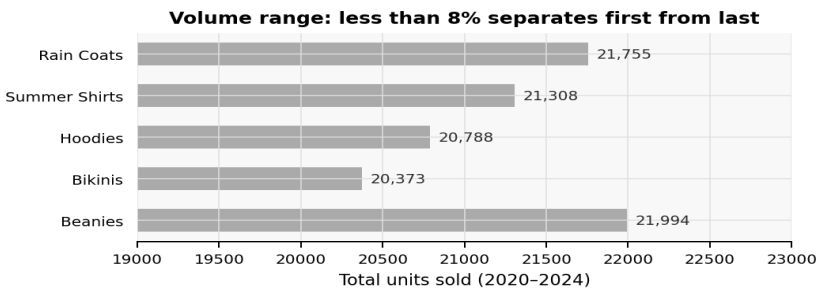


Figure 9 — Five-year unit totals (2020–2024). Under 8% separates highest from lowest.

The composite ranking tells a different story. Each product is ranked 1 to 5 across six dimensions (units sold, gross margin, inventory turnover, capacity utilization, customer satisfaction, items returned) and those ranks are averaged. Bikinis, Hoodies, and Rain Coats all land at 2.83 and form a clustered group of leaders. Summer Shirts sits at 3.00. Beanies is last at 3.50, pulled down by the weakest margin in the portfolio (29.0%) and the lowest customer satisfaction (8.41). Its satisfaction gap has held for 20 years. That is not a service problem or a logistics problem. It is a product problem.

Composite 6-metric ranking: 2020–2024

Product	Avg Rank	Margin	Satisfaction	Returns
Bikinis	2.83 ★	30.0%	8.51	1,124
Hoodies	2.83 ★	29.6%	8.43	1,078
Rain Coats	2.83 ★	29.8%	8.48	1,216
Summer Shirts	3.00	30.1%	8.53	1,180
Beanies	3.50 ✕	29.0%	8.41	1,177

Figure 10 — Composite 6-metric ranking (2020–2024). Clustered leaders at 2.83; Beanies weakest at 3.50.

On efficiency specifically: Rain Coats leads on inventory turnover at 5.68 turns per year, while Summer Shirts lags at 5.40. On capacity utilization, Hoodies runs the highest at 75.9% and Beanies the lowest at 72.6%. Neither spread is dramatic, but Beanies sitting at the bottom of both turnover rank and utilization rank reinforces the composite picture: it moves the most units but extracts the least operational value per unit of capacity committed. Yearly trend charts for all four Step 2 metrics (units, margin, satisfaction, turnover) and capacity utilization are included in the Appendix as Figures A1–A5.

Step 4: Correlation Analysis

Three operational pairs were tested across all pooled product-month observations:

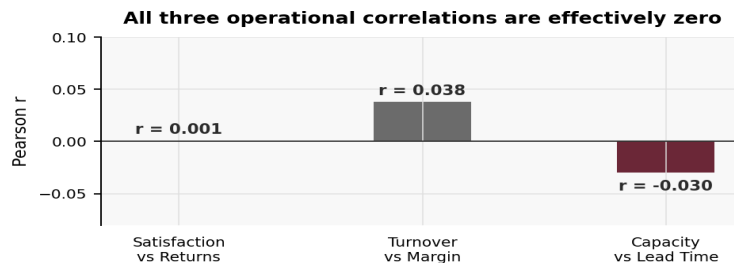


Figure 11 — Pooled Pearson correlations for three operational pairs. All near zero.

Pair	Pearson r	What it rules out
Satisfaction vs. Returns	0.001	Satisfaction scores do not predict return rates across the portfolio
Inventory Turnover vs. Margin	0.038	Faster turns do not reliably lift margins
Capacity Utilization vs. Lead Time	-0.030	Running lines harder does not shorten lead times

None of these pairs moves together in any linear way at the portfolio level. The practical consequence: there is no single company-wide dial to turn. What drives outcomes in Beanies is probably not what drives outcomes in Rain Coats. Improvement programs need to operate at the product level, not the aggregate.

Step 5: Strategic Recommendations

Bikinis is the right expansion candidate. It leads the composite ranking alongside Hoodies and Rain Coats, with consistent margin, satisfaction, and returns performance. Among the three tied leaders, Bikinis is the natural first choice given the swimwear context of this expansion.

Beanies needs a product review before any scaling decision. Last on satisfaction, last on margin, despite leading on volume. A 20-year satisfaction gap is not fixed with a service campaign. Root-cause analysis on materials, sizing, and return reasons should come first.

KPI targets need to operate at product level, not company-wide. The near-zero operational correlations confirm that aggregate targets hide product-specific problems. This matters more, not less, once new seasonal categories with distinct demand profiles are added to the portfolio.

The core finding across both analyses: volume is the wrong primary metric for this portfolio. Margin, efficiency, and customer satisfaction are where the decisions live, and those require product-level attention to be actionable.

Appendix

The combined Google Trends dataset is available in the submitted Excel file (Part1_Combined_Data sheet).

The Python script (build_analysis_package_xlsx.py) is submitted alongside this report because the Excel file contains no formulas: all values are computed outputs written programmatically. The composite ranking methodology (avg_rank_six_dims), Pearson correlations, five-year aggregates, and summary statistics are all calculated in the script from the raw source data before being written to the Excel sheets. The script is the only artifact that makes the quantitative methodology fully auditable and reproducible.

— End of Report —